WEBSITE DEVELOPMENT / DOCUMENT 08 — ANSWERS

Database & API Specifications

Completed questionnaire: what data is stored, where, and the one enquiry endpoint

PROJECT / WEBSITE	Radiant Institute of Professional Studies (RIPS), Sheikhupura — rips.edu.pk
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DATE / VERSION	17 October 2026 / v1.0 draft answers — for review
COVERAGE	42 questions · 8 sections · 3 entity specifications + content summary · 1 endpoint specification
INPUTS	Doc 01 PVS answers (draft v0.1) · Doc 02 PRD answers (draft v1.0) · Doc 03 SRS answers (draft v1.0) · Doc 07 architecture (ADR-01, ADR-05)

How to read this. References use the earlier documents' IDs: **PVS Q**n (Doc 01), **PRD Q**n (Doc 02), SRS IDs such as **B-01**, features **FEAT-001…006** and open decisions **OD-01…18** (Doc 03). **Not Decided** names its owner. **Not Applicable** gives a reason. *(proposed)* marks a recommendation the approver must confirm. Under the proposed architecture (Doc 07, ADR-01) the website owns **no database**: content is stored as files and enquiries in an institute-owned Google Sheet. The questions still apply to that data and to the one enquiry endpoint, so they are answered rather than skipped.

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A. Applicability & ownership

01. Which approved features require persistent structured data, and why can they not use static content or an existing service?

> **Enquiries (FEAT-003, FEAT-006)** are the only data written by visitors, and they need statuses, notes, export and deletion (SRS G-02, M-04). They **do** use an existing service: a Google Sheet owned by the institute (ADR-05, Doc 07). **Content** (programs, intakes, fees, notices, news) is structured but changes only when staff publish, so it is stored as files in the Git repository and turned into static pages. No feature needs a website-owned database.

02. Does the website own a database, consume an external API, expose its own API, or some combination?

> Combination, all small: **no website-owned database** (**Not Applicable** — content is files, enquiries live in Google Sheets). **Consumes** external APIs: Google Sheets (append and look up rows), email sending, spam-check verification. **Exposes** one endpoint of its own: the enquiry submission function, used only by the site's own form.

03. Which data and interfaces are in scope for the first release, and which are deferred?

> **Release 1:** content model (Program, Intake, Fee schedule, Notice, News post, Faculty member, Page, Site settings); Enquiry record; endpoint POST /api/v1/enquiry; integrations with Sheets, email and spam check. **Deferred:** applications with documents, Open Day bookings, student portal and payments (phase 2–3), which would justify a real database and authenticated API.

04. Who owns each dataset and API contract, including external dependencies?

Dataset / contract	Owner
Content files	Content owners per Doc 05 Q13; developer owns the schema
Enquiry sheet	Admissions officer (data); developer (columns); institute Google account
Enquiry endpoint contract	Developer
Google, email, spam-check contracts	Institute accounts; developer manages (SRS J-08)

05. What open product or policy decisions must be settled before a schema or endpoint can be final?

OD-06/ADR-05 (form handling), OD-05 (editing tool, which shapes the content file format), OD-07 (program codes and titles), OD-10 (retention periods), OD-15 (legal review: may enquiry data sit in Google?), and whether an enquiry reference number is shown to visitors (PRD Q80, Could).

B. Entities, fields & relationships

06. What real-world items must be stored, and what is the purpose of each entity?

Entity	Purpose	Stored in
Enquiry	A visitor's request to be contacted, and its follow-up	Google Sheet
Program	A study program and its key facts	Content file
Intake	Admission window for a session	Content file
Fee schedule	Approved fees for a program and session	Content file + PDF
Notice	Official announcement or merit list	Content file + PDF
News post, Faculty member, Page, Site settings	Public information	Content files
Staff user	Who can edit	Git provider and Google accounts (not stored by the site)

07. What fields does each entity need, with type, allowed values, required status and example value?

Given in the Entity specifications at the end: Enquiry, Notice and Program in full; the other content entities in a summary table. Examples are fictional.

08. Which fields are unique, which are identifiers, and which values may change over time?

Identifiers: enquiry_id; program slug/code; notice slug; news slug. Unique: submission_token (7 days). Never change after creation: enquiry_id, received_at, the visitor's original answers (except deletion on request), program code. May change: enquiry status and notes; program title, fee, status; notice file (replace) and status.

09. How do entities relate, and what happens to related records when one is changed or removed?

- Enquiry → Program by code. If a program is withdrawn, past enquiries keep the code and show "(withdrawn)".
- Notice → Program (or all). Withdrawn program: its notices stay until their archive date.
- Fee schedule → Program and Intake. Changes create a new version; old versions stay in Git history.
- Intake → Programs. Deleting an intake is not allowed while programs reference it; it is archived instead.
- Program codes are never reused, so old links and enquiries stay correct.

10. Which statuses and transitions are valid, and who can trigger each transition?

As SRS G-02 to G-04: Enquiry (New → Contacted → Follow-up → Visited/Applied → Enrolled; any → Not interested or Spam/Duplicate; not Enrolled → New) by admissions or administrator. Content (Draft → In review → Published → Archived) by the role owner; Archived automatically by date. Program status by content editor with Principal approval.

11. How will versions, audit history, timestamps and the source of a change be represented?

Content: every change is a Git commit with author, time and message, so full version history and restore come free. Enquiries: updated_by and updated_at columns plus Google Sheets' own version history; status notes kept as dated text. All times stored in Pakistan time with the offset (+05:00).

C. Data rules & lifecycle

12. Where is each value validated: user interface, server, database or external provider?

Enquiry: in the browser for quick feedback (SRS F-06) and **again in the function**, which is authoritative; the sheet's dropdowns restrict status values. Content: by the CMS field rules and again at build time (the build fails on a missing required field or unknown program code). Spam check: by the external provider, verified in the function.

13. What business rules must hold even if two requests arrive at the same time?

One submission token creates at most one enquiry. Because a sheet append is not a transaction, two identical requests arriving within milliseconds could rarely both be written; the duplicate flag (same mobile and program within 7 days) catches this and admissions marks one Spam/Duplicate. Content publishing is serialised by Git, so two publishes cannot overwrite each other silently.

14. Which records may be edited, corrected, archived or deleted, and who authorizes each action?

Record	Edit / correct	Archive	Delete
Enquiry	Status and notes: admissions. Visitor answers: only corrected on the visitor's request	—	Personal fields on request (admissions); whole rows after retention (administrator)
Notice, merit list	Role owner; merit lists with Principal's approval	Automatic by date	Administrator, with Director's approval
Program, fees, intake	Content editor / admissions with approval	Program set to Withdrawn	Not deleted; kept in history

15. What retention, deletion, export and backup rules apply to each data class?

> As SRS M-05 and M-06: enquiries 12 months after the intake (proposed, OD-10), then deleted or reduced to program, city and status; personal fields wiped within 14 days of a request; exports only by admissions/administrator, logged, deleted from devices after use; content kept in Git permanently, notices removed from the site after 2 years; backups weekly, 14 days kept.

16. How will duplicates, missing values, legacy data and migrations be detected and resolved?

> Duplicates: flag rule above. Missing values: required fields rejected by the function; content with missing required fields fails the build. Legacy data: none migrated (SRS M-09). Content migration from the old site is manual and checked against the content checklist (SRS M-11).

17. What test data can be used without exposing real people's information?

> Fictional enquiries ("Test Student One", numbers 0300 0000001–0300 0000099, emails @example.com), sample PDFs marked SAMPLE, a staging sheet separate from production (Doc 07 Q33). Never copies of real enquiries or real merit lists.

D. Privacy & access

18. What personal or sensitive fields exist, and what is the reason for collecting each one?

Field	Reason
Full name	To address the person when calling back
Mobile	To call or WhatsApp back — the main purpose
Email (optional)	Auto-reply and written follow-up
City	To advise on travel and plan outreach
Qualification, marks (optional)	To advise on eligibility
Message (optional)	To answer the specific question
Merit lists (names in PDFs)	Official publication of admission results

No CNIC, B-form, date of birth or address is collected in release 1.

19. For every role, which records and fields can be read, created, updated, deleted or exported?

Role	Enquiries	Content
Visitor	Create only (own submission); cannot read anything back	Read published
Admissions officer	Read, update status/notes, delete personal fields on request, export	Create/update notices, merit lists, fees, dates
Content editor	None	Create/update programs text, news, faculty, general notices
Administrator	All, including whole-row deletion	All
Enquiry function	Append only; reads recent rows only to check token and duplicates	None

20. How is record ownership checked on the server, including direct access to another record's URL or ID?

> There are no per-visitor records to protect: the endpoint never returns stored data and accepts no record IDs, so another person's enquiry cannot be requested by URL. Enquiry rows are reachable only inside the institute's Google account, governed by sheet sharing. The function's Google credential can only reach the enquiry sheet.

21. What should be encrypted, masked or omitted from logs, analytics, backups and error messages?

> Omitted from logs and error messages: names, numbers, emails, marks, messages, tokens and keys — logs keep enquiry_id, outcome and error type only. Omitted from analytics: everything in the form (SRS R-08). Masked in admissions alert emails: nothing (admissions needs the details) but emails are sent only to the institute mailbox. Backups and exports: stored only in institute storage, encrypted at rest by the provider.

22. How are consent, access requests and deletion requests handled if applicable?

> Consent: required checkbox; consent_version recorded with each enquiry so wording changes are traceable. Access, correction and deletion requests: handled by admissions as in Doc 04 UC-03 within 14 days (proposed); deleted_on_request_at recorded; logged in the activity record.

E. API resources & behavior

23. What resource or action does each endpoint represent, and which journey calls it?

> One endpoint: **POST /api/v1/enquiry** — "create an enquiry", called by the enquiry and contact forms (J1, UC-01). There are no read, update or delete endpoints; staff work in the sheet and the CMS, which are external services.

24. What method, path, request fields and response fields does each endpoint require?

> See the API endpoint specification at the end. Request: JSON with the enquiry fields, consent, submission token, spam-check token and honeypot. Success response: 201 with {"status":"received"} (and a reference number if Could item PRD Q80 is approved).

25. What authentication and authorization rule applies before any data is returned or changed?

> No user authentication (public form). Before writing: request must come from the site's own origin; spam-check token verified with the provider; honeypot empty; rate limit not exceeded; all fields valid. Nothing is ever returned except success or an error code.

26. What validation errors, business errors and HTTP status codes should callers expect?

> Listed in the endpoint specification: 201, 200 (repeat token), 400, 403, 405, 413, 415, 429, 500, 503. Validation errors return field codes (e.g. mobile_format) that the page maps to the SRS F-07 messages, so wording is changed in one place.

27. How are lists filtered, sorted and paginated, and what limits protect the service?

> The endpoint returns no lists: **Not Applicable**. Public lists (programs, notices) are built into static pages with filtering in the browser (Doc 05 Q18). Limits protecting the endpoint: body ≤ 8 KB, 5 requests per hour per device/IP, 10-second upstream timeout.

28. Which operations must be idempotent, and how are retries and duplicate submissions handled?

> Create enquiry is made idempotent by the submission token: a repeat returns the same success and writes nothing. The browser may retry after a failure with the same token. The function retries the sheet write once internally.

F. Contracts & integration edges

29. What API versioning and compatibility rule will protect existing clients?

> The only client is the site's own form, deployed together with the function. Still, the path carries v1; a breaking change gets /v2 and the old path is kept until cached pages have expired (at least 7 days).

30. What timeouts, rate limits, quotas and retry rules apply to each external integration?

Integration	Timeout	Retry	Quota watch
Google Sheets	10 s	Once	Provider limits; alert at 80% of any daily quota
Email sending	10 s	Once; enquiry stored anyway	Monthly send limit of chosen plan
Spam check	5 s	None; fall back to honeypot + rate limit	Free-tier limit

31. What happens when an upstream service is unavailable or returns partial or unexpected data?

> Sheet unavailable after retry: return 503, send the enquiry to admissions by email so it is not lost, and alert the administrator. Email unavailable: row saved, response still 201, administrator alerted. Unexpected or partial reply: treated as failure, never reported as success (SRS Q-07).

32. Do callbacks or webhooks need signatures, replay protection and a delivery-status log?

> **Not Applicable** in release 1: no inbound webhooks. If the CMS or host sends build notifications later, they will use the provider's signing secret.

33. What files or large payloads are exchanged, and how are upload, download and access controlled?

> The endpoint accepts no files. PDFs and images are uploaded by staff through the CMS into the repository and served publicly as static files; merit-list PDFs carry a no-index header (Doc 07 Q18). Phase 2 applications with documents will need a separate, access-controlled storage design.

G. Performance & operations

34. Which queries or endpoints may handle large volumes, and what measured target do they need?

> Only the enquiry endpoint at admission peaks: target 95% of submissions complete in under 5 s (SRS P-04) at up to 1 submission per second. Public content is static, so there are no runtime queries.

35. What indexes or caching might help, and what will be measured before adding them?

> No database indexes (**Not Applicable**). The token and duplicate check reads only recent rows; measure function duration and sheet size each month, and move the token check to a key-value store or the store to a database if duration passes 3 s or rows pass about 20,000 a year.

36. How will schema changes be deployed and rolled back while existing data remains usable?

> Content schema: CMS configuration, templates and any one-off conversion of existing files are changed in one reviewed commit; the build validates every file; rollback is reverting the commit. Sheet: new columns added at the right, never renaming or reordering existing ones; the function writes by column name so old rows stay valid.

37. What monitoring reveals slow queries, failed requests and unusual access without logging sensitive content?

> Function logs with duration, status code and error type; alerts on 3 failures per hour and on unusual volume (more than 50 submissions an hour). Google account activity alerts for the sheet; activity log for exports (SRS L-10). No field values in any log.

38. How will backups and restores be tested, including who can perform a restore?

> Weekly export of the sheet and mirror of the repository (Doc 07 Q17). Restore drill before launch and every 6 months into staging, by the administrator, witnessed by the Director; result recorded. Only the administrator can restore.

H. Verification & sign-off

39. What example records and requests demonstrate every required field and relationship?

> A sample set on staging: 5 programs (one Withdrawn), 1 intake, 2 fee schedules (old and new), 6 notices (one of each type, one archived, one merit list), and 20 fictional enquiries covering every status, a duplicate, a deletion on request, and an enquiry for “not-sure”. Example request and responses are in the endpoint specification.

40. Which permission-denied, invalid-input, concurrency and retry cases must contract tests cover?

> - Each invalid field returns 400 with the right code.
> - Honeypot filled and bad spam token return 403.
> - Same token twice: one row, two 200/201 responses.
> - Two simultaneous identical requests: at most a flagged duplicate.
> - Sixth request in an hour returns 429.
> - Sheet down returns 503 and an email reaches admissions.
> - Formula-looking input (=SUM…) is stored as plain text.
> - Content editor cannot publish a merit list (CMS check).

41. How will consumers review example responses and confirm that no private fields leak?

> The developer shares the example responses with the Director and admissions officer; all success and error bodies are checked to contain no stored data, and logs from the test run are searched for the test names and numbers (none should appear).

42. What evidence establishes that the database and API specification matches the approved workflows?

> A traceability table: each field and endpoint case → SRS requirement (F-02, G-02, M-04, Q-03) → journey/use case (J1, J6, J8, UC-01, UC-03) → passing test. Approved by the approver (OD-01) with the contract-test report.

Entity specification — Enquiry

Entity / purpose	Enquiry — a visitor's request to be contacted, and its follow-up
Source requirement / owner	FEAT-003, FEAT-006 · SRS F-02, G-02 · admissions officer (data), developer (columns)
Primary key / unique rules	enquiry_id; submission_token unique for 7 days
Retention / deletion rule	12 months after intake (proposed, OD-10); personal fields wiped on request within 14 days

Field / type	Required? / allowed values	Example / default	Sensitive? / access	Validation / index
enquiry_id · text	Yes · ENQ-YYYY-nnnnnn, generated	ENQ-2026-000123	No · admissions, admin	Unique
received_at · date-time (PKT)	Yes · set by function	2026-10-17T14:30:00+05:00	No	Sort key
full_name · text	Yes · 3–60 chars	Test Student One	Personal · admissions, admin	SRS F-03
mobile · text	Yes · stored as +923XXXXXXXXX	+923000000001	Personal	Normalised; search key
email · text	No · ≤100 chars	test1@example.com	Personal	Email format
program_code · text	Yes · active program code or not-sure	pharm-d	No	Must exist in program list
city · text	Yes · list value or free text ≤40	Sheikhupura	Personal (low)	—
qualification · text	No · Matric/O-level, FSc Pre-Medical, FSc Pre-Engineering, ICS, FA, ICom, A-level, Other	FSc Pre-Medical	No	Enum
marks_percent · number	No · 0–100, 1 decimal	78.5	Sensitive	Range
message · text	No · ≤500 chars	When do classes start?	Personal	Length; formula-escaped
consent · yes/no + consent_version	Yes · must be yes	yes · v1	No	Must be yes
source_page, campaign · text	No · page path; UTM tag	/programs/pharm-d · fb-oct	No	No personal data allowed
submission_token · text	Yes · one-time value from the form	(random)	No	Unique for 7 days
status · text	Yes · New, Contacted, Follow-up, Visited/Applied, Enrolled, Not interested, Spam/Duplicate	New	No · admissions edit	Dropdown
possible_duplicate · yes/no	Yes · set by function	no	No	Same mobile + program ≤7 days
notes · text	No · staff only	Called 17 Oct, will visit	Personal	—
updated_by, updated_at	Yes when changed	admissions@ · 2026-10-18 10:05	No	Sheet history also records
deleted_on_request_at · date	No	2026-11-01	No	Set when personal fields are wiped

Relationships, transitions, audit, concurrency	program_code → Program. Status transitions per SRS G-03. Audit: updated_by/at + sheet version history. Concurrency: token check + duplicate flag (Q13).
Migration and acceptance examples	No migration. Example: Test Student One, +923000000001, pharm-d, Sheikhupura, consent yes → status New, possible_duplicate no.

Entity specification — Notice

Entity / purpose	Notice — official announcement or merit list
Source requirement / owner	FEAT-004 · SRS F-02, L-04 · admissions officer
Primary key / unique rules	slug; one merit list per program + session + number
Retention / deletion rule	Archived on archive_on; removed from site after 2 years; kept in Git

Field / type	Required? / allowed values	Example / default	Sensitive? / access	Validation
slug · text	Yes · lowercase-hyphen, unique	2026-27-pharm-d-merit-list-1	No	Unique; becomes the URL
title · text	Yes · 5–120 chars	Pharm-D — Merit List 1, 2026–27	No	Length
type · text	Yes · Admission, Merit list, Exam, General	Merit list	No	Enum; merit list limited to admissions role
program · reference	Yes · program slug or all	pharm-d	No	Must exist
session · text	Required for merit lists · YYYY–YY	2026–27	No	Pattern
publish_at · date-time	Yes · default now	2026-10-17 15:00	No	—
archive_on · date	Yes · default intake close + 30 days	2026-12-15	No	After publish_at
file · PDF	Yes · ≤5 MB	merit-list-1.pdf	Merit lists contain student names	Type and size
no_index · yes/no	Auto yes for merit lists	yes	—	Set by build
status · text	Yes · Draft, In review, Published, Archived	Draft	No	Workflow
approved_by · text	Required for merit lists	Principal	No	Not empty

Relationships, transitions, audit	program → Program. Draft → In review → Published → Archived. Every change is a Git commit.
Acceptance examples	Merit list without session or approver fails validation; build marks merit lists no-index.

Entity specification — Program

Field / type	Required? / allowed values	Example	Validation
slug / code · text	Yes · stable, never reused	pharm-d	Unique; used by enquiries and notices
title · text	Yes	Pharm-D (Doctor of Pharmacy)	Titles pending OD-07
field · text	Yes · Pharmacy, Physical therapy, Nursing, Allied health, Computing, Business	Pharmacy	Enum
level · text	Yes · Undergraduate, Postgraduate, Short course	Undergraduate	Enum
duration · text	Yes	5 years	—
eligibility · rich text	Yes	FSc Pre-Medical, minimum marks to confirm	Approved by Principal
fee_per_year_pkr · whole number	No (shows “contact admissions” if empty)	Sample only	≥ 0
affiliation · text	Yes	Wording pending OD-08	Approved by Director
status · text	Yes · Active, Not offered this intake, Withdrawn	Active	Enum
overview, outline, careers · rich text	Overview yes; others no	—	—
image · file	No	pharmacy-lab.webp	≤5 MB before compression
last_updated, approved_by	Yes	2026-10-17 · Principal	—

Other content entities (summary)

Content entity	Key fields	Relationships
Intake	session, opens_on, closes_on, programs[]	→ Program (many); drives status badges
Fee schedule	session, program, lines[label, amount_pkr, per], pdf, approved_by, approved_on	→ Program, Intake; old versions kept in Git
News post	slug, title, date, summary, body, image, status	Optional → Program
Faculty member	name, designation, qualification, department, photo, order	Optional → Program field
Page	slug, title, body, last_updated	—
Site settings	phone, WhatsApp, email, address, office hours, map link, social links	Used on every page and in auto-replies

API endpoint specification — Create enquiry

Method / path / version	POST /api/v1/enquiry (path proposed; depends on ADR-05)
Related workflow / owner	J1 · UC-01 · FEAT-003 · developer
Authentication / role	None (public form); same-origin check, spam-check token, honeypot, rate limit
Record-level permission rule	Create only; returns no stored data; accepts no record IDs

Request body (JSON)

Field	Example	Notes
full_name	"Test Student One"	Required
mobile	"0300 0000001"	Required; normalised to +923000000001
email	"test1@example.com"	Optional
program_code	"pharm-d"	Required
city	"Sheikhupura"	Required
qualification	"FSc Pre-Medical"	Optional
marks_percent	78.5	Optional
message	"When do classes start?"	Optional
consent	true	Required, must be true
submission_token	"(random)"	Required; one-time
captcha_token	"(from widget)"	Required unless spam service down
website	""	Honeypot; must be empty
source_page, campaign	"/programs/pharm-d", "fb-oct"	Optional

Success: status, response, side effects	201 {"status":"received"}. Side effects: row appended (status New, duplicate flag), email to admissions, auto-reply if email given.

Case	Status / error code	Caller action or recovery
Field invalid	400 · validation_failed + list of {field, code}	Show SRS F-07 messages beside fields
Spam check failed or honeypot filled	403 · rejected (generic)	Show generic failure; no detail
Token already used	200 · same success body	Show Thank-you (idempotent)
Body too large (> 8 KB)	413 · too_large	Show generic failure
Wrong content type or method	415 / 405	Developer error
Rate limit (5 per hour per device/IP)	429 · rate_limited + Retry-After	“Too many attempts — please WhatsApp us”
Sheet or email unavailable	503 · upstream_unavailable	Failure message with WhatsApp and phone; values kept
Unexpected error	500 · internal (no details)	Same as 503; administrator alerted

Pagination, limits, idempotency, timeout, privacy	No pagination. Body ≤ 8 KB; 5 per hour per device/IP; idempotent by submission_token; 10 s upstream timeout, browser gives up at 15 s; no personal data in logs or responses; spreadsheet-formula characters escaped.
Contract tests and decision status	Tests listed in Q40. Status: Proposed — final once ADR-05/OD-06 is decided.